Agency-Enhanced Visual Search in VR: Robust to Distraction, Delay, and Perspective Shifts

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Abstract

The sense of agency, the internal feeling of controlling one's actions and their outcomes, fundamentally shapes user interaction in virtual environments. Although extensively studied for its subjective impact, agency's implicit yet critical role in guiding visual-spatial attention has been largely overlooked. This study investigated whether the sense of agency, conferred by prior active control, enhances subsequent visual search efficiency for the previously controlled stimulus under conditions of attentional, temporal, and spatial perturbation. Our results show that agency-driven attentional benefits are remarkably robust, persisting despite competing salient visual distractors, delayed outcomes, and changes in spatial layout. Furthermore, when a delay was introduced between the control and visual search, the agency effect attenuated for targets presented beyond the operators' peripersonal space. These findings provide valuable insights for constructing immersive user experiences and advancing theoretical frameworks in human-computer interaction, suggesting efficient strategies to support sustained user engagement in virtual and augmented reality.

CCS Concepts

• **Applied computing**; • **Law, social and behavioral sciences**; • **Psychology**;

Keywords

Sense of agency, Visual search, Virtual reality, Peripersonal space

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1 Introduction

The satisfying sensation of pressing a key and seeing a letter instantly appear on the screen epitomizes the sense of agency, the fundamental feeling of influencing one's environment through one's own actions [1-3]. This intuitive connection between action and outcome is a cornerstone of effective human-computer interaction (HCI) [4-7]. Within HCI, researchers have extensively explored the sense of agency from various perspectives, including autonomy in human-centered design [8-11], dynamics in automation and joint action with co-agents, and its application as a metric for interface assessment [12-15]. A common thread across this body of work is the treatment of agency primarily as an explicit subjective experience. In contrast, cognitive neuroscientists have studied agency by revealing its implicit mechanisms [16-18]. For example, sensorimotor predictions from voluntary actions (e.g., moving a finger) produce sensory attenuation, reducing the perceived intensity of subsequent touch by that finger. Further research aligning with this approach extends the agency effects induced from self-generated actions to tool-mediated actions [3].

2 Related work

2.1 Implicit-Explicit Distinction in Sense of Agency

Researchers have recently proposed a two-step framework differentiating implicit and explicit dimensions of the sense of agency [19, 20]. At the implicit level, a low-level, non-conceptualFeeling of Agencyarises, which involves the pre-reflective tagging of actions as self-caused or not. Although potentially conscious, this feeling does not entail reflective awareness and is thought to originate from sensorimotor processes such as efferent signals and sensory feedback. At the explicit level, a higher-order, conceptualJudgment of Agencyis formed. This reflective process involves attributing agency to oneself or others and is influenced by higher-order cognitive sources such as prior beliefs, social context, and situational cues.

Consequently, researchers have developed diverse methodologies to quantify the sense of agency [20-22]. Researchers in HCI frequently employ direct self-report instruments, such as questionnaires or rating scales, that explicitly ask participants to indicate how much control they perceive over an event or to what extent they believe their actions caused a particular outcome. Explicit measures of agency, however, are limited by cognitive biases and

may not accurately reflect implicit processes. For instance, individuals are usually prone to a self-attribution bias, often over-claiming responsibility for outcomes [2]. Furthermore, subjective personal criteria for causation vary across individuals: a liberal criterion can inflate perceived agency, while a conservative one can result in an underestimation [5].

Alternatively, cognitive scientists have adopted paradigms like intentional binding and sensory attenuation to detect and evaluate the implicit sense of agency. Specifically, the intentional binding paradigm requires participants to perform an action that causes a subsequent outcome (e.g., a sound) and then estimate the timing of both events. Researchers have uncovered a heightened sense of agency from the binding effect, a perceptual phenomenon whereby people perceive the interval between their action and its effect as shortened [16]. Conversely, the sensory attenuation paradigm leverages the established finding that the brain's internal predictive models dampen the sensory processing of self-generated actions [17, 23]. The degree of this perceptual dampening serves as an implicit metric for the strength of an individual's sense of agency. Within applied contexts, intentional binding serves as a primary tool for assessing the impact of input interfaces on agency [12-15], while sensory attenuation is employed to quantify agency-induced shifts in perceptual sensitivity, most commonly in response to auditory [3, 18] and visual stimuli [24-26].

2.2 Agency Modulates Visual Attention

Visual attention is essential for rapidly selecting important environmental information. The sense of agency modulates attention, producing effects that include facilitation but are also bounded by specific constraints. Induced through methods like directional control, agency improves visual search efficiency (i.e., faster reaction time, higher accuracy) when the active control is reliable [27, 28], with benefits scaling monotonically with control probability [29]. A "self pop-out" effect also occurs, where self-controlled avatars are detected faster than passive ones, especially in large arrays [30]. Detection is likewise quicker for self-generated versus observed pre-recorded motion [31]. These advantages, however, are diminished by competing salient distractors [32].

Crucially, these attentional enhancements appear to be driven by implicit sense of agency rather than by explicit intention or subjective judgment. For instance, performance benefits disappeared when participants had a false sense of control, indicating that implicit sensorimotor processes, not conscious attribution, underlie agency-based attentional modulation [33]. This finding supports the idea that attentional effects may serve as a viable implicit measure of agency.

Collectively, the extant literature converges on the principle that agency prioritizes attention toward self-controlled objects, an effect strongest under reliable control but attenuated by salient attentional distractions. Neurophysiological support of this view comes from an electroencephalography study examining the perceptual prioritization of objects associated with agency [31]. Specifically, participants controlled two groups of shapes via a cursor, with one group designed to be more responsive. Although task performance did not require awareness of the controllability gradient, steady-state visual evoked potential (SSVEP) amplitudes were significantly higher in the more responsive group. This neural signature confirms that agency automatically upregulates attentional processing of controlled objects, even when control is task irrelevant.

2.3 Temporal and Spatial Adaptability of Agency

In human-computer interaction, immediate feedback following a user's input is usually expected, as it reinforces the sense of agency by confirming that the system's response is a direct consequence of their control. Real-world interactions, however, are often characterized by delays between input and feedback, which may disrupt the established sense of agency. The empirical literature suggests agency is indeed vulnerable to temporal perturbation. For instance, intentional binding, an implicit measure of agency, decays over intervals of seconds [16, 34], showing only partial resilience to delays up to 4 seconds [35]. Similarly, delayed action consequences attenuate the sensory dampening that signals the sense of agency [18, 23]. Collectively, current evidences underscore that the sense of agency, while resilient to brief interruptions, is significantly compromised by prolonged delays.

Consider another scenairo when a drone operator navigating a complex environment. While hovering, the operator focuses on a specific target using an egocentric frame of reference, exercising precise control. As the drone moves forward through a narrow passage, the operator must shift to an allocentric, trajectory-based frame to avoid obstacles. This necessary transition between spatial reference frames, all while maintaining operational control, prompts a key question: can the sense of agency, established in one spatial context, facilitate performance in another? If so, what cognitive mechanisms support this flexibility?

3 Current Study

Prior work reveals a cross-disciplinary methodological divergence between HCI and cognitive science in studying agency, underscoring the need for a synthetic framework. This raises a critical research question: can implicit measures of agency be validated as objective metrics for evaluating and designing interactive systems?

To address this gap, we conducted three controlled VR experiments. We operationalized the implicit sense of agency as self-generated active control over a stimulus and measured its effect on visual search efficiency for that stimulus. By introducing three distinct types of disruption, exogenous distraction, temporal delays, and spatial layout shifts, within a unified paradigm, we examined the robustness of agency-driven attention. Our study investigated the following research questions:

RQ1: How do agency-based attentional benefits compete with salient exogenous cues?

RQ2: Does the agency-induced attentional guidance withstand a temporal delay?

RQ3: Does this benefit persist when the spatial configuration of stimuli changes?

This experimental approach bridges precise, laboratory-based implicit measures with the complex dynamics of immersive environments. Through quantitative analysis, we established agency as a critical and quantifiable implicit factor. By shifting the focus from subjective phenomenology to objective measurement, this work

lays a foundation for optimizing interactive systems informed by implicit human experience.

4 Experiment 1: Agency against Attentional Distraction

Visually salient but task-irrelevant elements in VR simulations can deplete attention needed for primary tasks. Experiment 1 investigated the resilience of agency-driven attention in the presence of salient exogenous cues, specifically, red-colored distractors that competed for attentional resources. To extend earlier 2D findings [32] into an immersive 3D context, we employed a virtual reality paradigm where participants controlled a virtual hand via rotational wrist movements. Critically, only agency-relevant stimuli were synchronized with participants' actions; uncontrolled stimuli were pre-recorded and exhibited no temporal alignment with user input. This design ensured a clear distinction between self-generated and external motion, strengthening the induction of agency while minimizing visual confounds. By omitting visual feedback of the user's real hand, we further isolated the sense of agency from effects of bodily illusion or visual incongruence.

4.1 Participants

Prior to conducting the experiments, we used G*Power [36] to estimate the required sample size. For a repeated-measures Analysis of Variance (ANOVA) with a 2 (Agency: Control target vs. Control distractor) × 2 (Cue validity) design, we anticipated a minimum sample size of 30 to detect a medium effect size (f = .25, $\eta^2 = .06$), with an alpha level of 0.05 and a power of 0.9. Taking a conservative approach, we recruited participants via snowball sampling at Beijing Normal-Hong Kong Baptist University. Initial data collection involved forty-six students. Seven participants were excluded due to motion sickness in VR or failure to follow instructions, resulting in a final sample of 39 (17 male, 22 female; Age: Mean ± SD: 20.44 + 0.64 years). All participants were right-handed and naïve to the study's purpose. All provided informed consent by reading and signing a digital form prior to the experiment. The study adhered to the local institutional ethical guideline and the ACM Publications Policy on Research Involving Human Participants and Subjects.

4.2 Stimuli and Procedure

Experiment 1 consisted of two main phases. The first phase, referred to as the Control Phase (see Figure 1), began with the presentation of a fixation cross for 500 ms, dividing the screen into four quadrants. After one second, four virtual hands appeared on the screen. Participants were instructed to control one of the virtual hands by rotating their wrists, thereby fostering a sense of agency. The hand they controlled moved in real-time in response to their actions, while the other three hands displayed pre-recorded rotational movements. Participants were then asked to identify which hand model they felt they had the most control over, emphasizing that this task was distinct from the subsequent phase of the experiment. The duration of the Control Phase was established following an exploratory preliminary study in which we tested five participants across six different control durations (ranging from 2 s to 3 s, with 200 ms increments). Our analysis revealed no significant differences in the induction of the sense of agency among these durations. To minimize participant fatigue, we ultimately selected a duration of 2 s for our formal experiment.

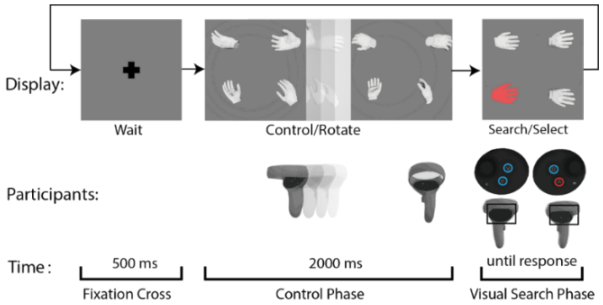

Figure 1: Illustration of a single trial in Experiment 1. Participants controlled a virtual hand that performed a rotation and then searched for a target hand among three distractors. In this specific trial, the salient cue incorrectly highlighted a distractor, while the correct target was located in the bottom-right quadrant.

Immediately following the Control Phase, four static hand models appeared in the Visual Search Phase, with only the target hand differing from the distractors. Participants had to quickly and accurately identify the target hand by pressing spatially congruent buttons on the controller. Reaction time was measured from the start of the Visual Search Phase to the button press. To introduce exogenous interference, the color tint of either the target hand (valid attentional cue) or one of the distractor hands (invalid attentional cue) was changed to red.

The experiment involved two factors, each with two levels: whether the virtual hand controlled by the participant in the initial action Control Phase was the search target in the Visual Search Phase (Control target) or a distractor (Control distractor), and whether the highlighted stimulus in the Visual Search Phase was the search target (valid attentional cue) or not (invalid attentional cue). Throughout the experiment, the display remained fixed to ensure that the rendered image did not track head movement, thus maintaining a consistent relative position with the eyes.

The minimum visual angle of one side of the hand model was 13.37°, whereas the maximum visual angle was 24.91°. The scene incorporates four placeholders that delineate the spatial positioning of the virtual hand. In configurations where the layout remained constant, the interspersed distance between each off-diagonal placeholder was equivalent, maintaining an FoV of 35.7°. Each placeholder exhibited a deviation of 18.3°, both horizontally and vertically, from the fixation cross.

Experiment 1 comprised a total of 74 trials, which included 10 practice trials followed by 64 experimental trials. The entire experiment took approximately ten minutes to complete.

4.3 Apparatus

We constructed the virtual environment using Unreal Engine and rendered it on an Oculus Quest 2 headset (Refresh rate: 120 Hz; Field of View: 89° ± 4° horizontal × 93° ± 5.1° vertical). We implemented the trial structure setting and collected data using the UXF framework (v.2.4.3) [37].

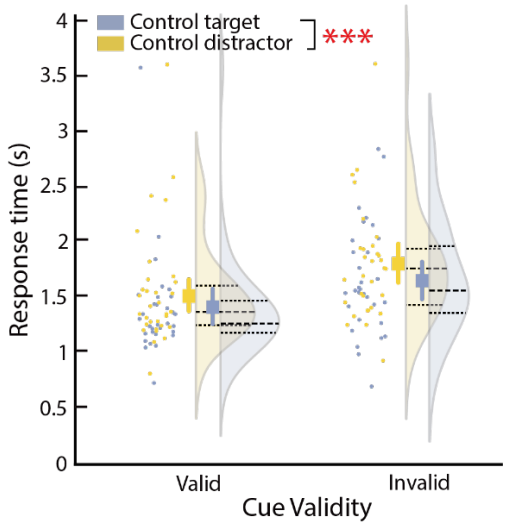

Figure 2: Participants were faster at finding the target when they had controlled it during the prior Control Phase, compared to when the target had not been controlled before. ***, p < .001

4.4 Data analysis

Trials with incorrect responses or outlying reaction times (exceeding ±2.5 SDs) were excluded to reduce noise from attentional lapses, resulting in a 9.6% exclusion rate (7.0% errors, 2.6% outliers). The implicit agency effect was calculated as the reduction in search time for the previously controlled target (Control target) versus the uncontrolled target (Control distractor) in the Control Phase.

4.5 Results and Discussion

During the initial Control Phase, all participants reported finding the controlled hand effortlessly. In the subsequent Visual Search Phase, they maintained a high and consistent level of accuracy (average 93%), with no significant differences between conditions. Therefore, our analysis focused primarily on response time for visual search task.

A 2 (Agency: Control target vs. Control distractor) × 2 (Cue validity: Valid vs. Invalid attentional cue) repeated measure ANOVA on response time revealed significant main effects of both agency ($F(1, 38) = 26.72$, $p < .001$, $\eta^2 = .41$), and cue validity ($F(1, 38) = 46.05$, $p < .001$, $\eta^2 = .54$, see Figure 2). These results indicate that response time was significantly faster both when the target had been under participant's control during the prior Control Phase and when a visually salient cue validly indicated the target location. Specifically, response time was 11.32% faster when the search target corresponded to the previously controlled virtual hand. However, no significant interaction was observed between cue validity and agency ($F(1, 38) = 1.31$, $p = .26$, $\eta^2 = .03$).

The combined results address RQ1 and indicate that the sense of agency, established through prior control, effectively directs attention and persists even when competing with visually salient distractions. Notably, the attentional benefit conferred by agency remained robust under exogenous cue interference. However, effect size comparisons revealed that the exogenous cue ($\eta^2 = .54$) exerted a stronger influence on visual search performance than agency ($\eta^2 = .41$), suggesting that bottom-up salience may have a more pronounced impact on attentional allocation than top-down agency-driven modulation. Importantly, we did not observe interaction effect between cue validity and agency, indicating that the facilitating effect of agency operates independently and remains intact despite the presence of external attentional competition.

Our VR results contrast with the earlier 2D findings of Vilanova-Goldstein and colleagues [32], who observed no agency effect alongside salient exogenous cues. This discrepancy likely stems from two critical enhancements in our paradigm: (1) the use of asynchronous movement onset to clearly signal which hand was under active, self-generated control versus a pre-recorded movement, and (2) the immersive VR context, which enables the integration of proprioceptive and visual information, thereby fostering a more robust sense of agency.

5 Experiment 2: Temporal Adaptability

In real-world applications, human-computer interactions are often disrupted by system delays. Given Experiment 1's finding that agency facilitates subsequent visual search, we ask: does this advantage withstand temporal interruptions? Experiment 2 directly tested the persistence of the agency effect by introducing a delay between the Control and Visual Search Phases. This design helps determine whether the sense of agency retains its attentional advantages across waiting periods, an essential feature of many real-world interactive systems.

5.1 Participants

For a 2 (Agency: Control target vs. Control distractor) × 4 (Delay intervals) repeated-measures design, we anticipated a sample size of 20 to detect a medium effect size ($f = .25$, $\eta^2 = .06$), with an alpha level of 0.05 and a power of 0.9. Taking a conservative approach, we recruited a new group of 31 participants via snowball sampling (12 males, 19 females, Age: Mean ± SD: 25.38 + 0.50 years) at the Hong Kong University of Science and Technology (GZ) for experiment 2. All participants were right-handed and naïve to the study's purpose. All provided informed consent by reading and signing a digital form prior to the experiment.

5.2 Stimuli and Procedure

In this experiment, we examined four conditions with varying delay intervals: no delay, 0.5 seconds, 1 second, and 1.5 seconds (see Figure 3). During the delay, all virtual hands were removed from view, leaving only the background visible. The general experimental procedure was consistent with that in Experiment 1. However, we modified the target feature in the Visual Search Phase to a hand with folded fingers. This adjustment was necessary because, when transitioning Experiment 2 to a new platform, pilot testing (N=5) revealed an unexpectedly low baseline accuracy compared to Experiment 1. Changing to a more distinctive folded finger configuration maintained comparable task difficulty across experiments. This preserved our core experimental design, an odd-one-out visual search

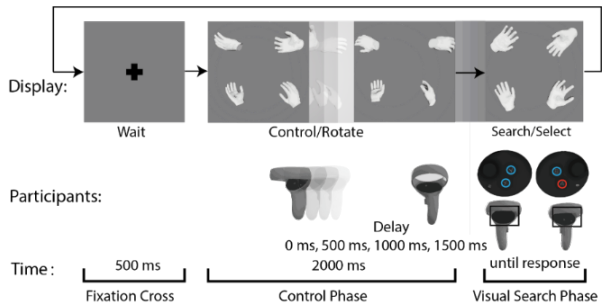

Figure 3: Illustration of single trial in Experiment 2. The target feature is a folded finger in the Visual Search Phase. During the delay, participants see blank background only.

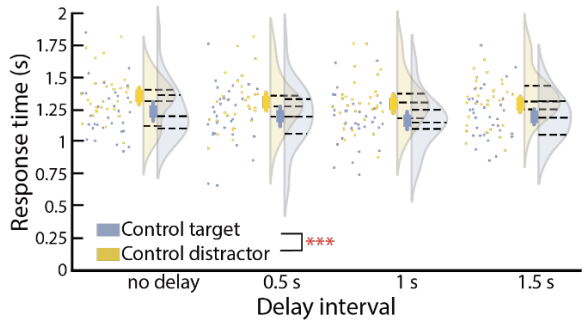

Figure 4: Across four delay interval conditions, participants identified targets significantly faster when they had previously controlled them compared to non-controlled targets. ***, p < .001.

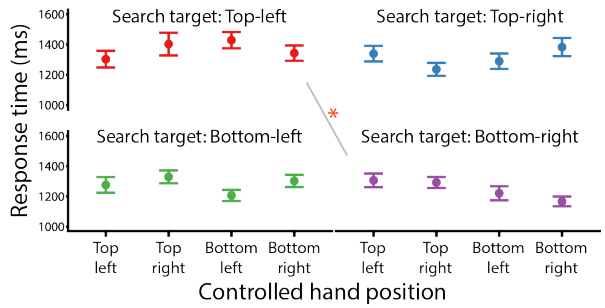

Figure 5: Illustration of response times as a function of search target position and the controlled hand position in four quadrants in Experiment 2. *: p < .05.

task, while ensuring participant performance remained at difficulty levels comparable to Experiment 1.

Experiment 2 consisted of 106 trials in total, including 10 practice trials and 96 experimental trials. The four delay conditions were randomly organized. Participants completed the entire experiment in approximately fifteen minutes.

5.3 Apparatus

We rendered Experiment 2 in an Oculus Quest 2 headset (refresh rate: 120 Hz, FoV: horizontal 89° (±4°) × vertical 93° (±5.1°). We used Unity Engine to develop and run all programs and directly streamed them to the headset using the Oculus home app. The remaining configuration was consistent with Experiment 1.

5.4 Data analysis

We excluded incorrect trials and outliers (RT < 30 ms or > 3000 ms) from the data analysis which comprised 1.9% of the data (1.5% errors, 0.4% outliers). Similar to Experiment 1, participants maintained a high and consistent level of accuracy (>90%), with no significant differences between conditions. Therefore, our analysis focused primarily on reaction time for the visual search task.

5.5 Results and Discussion

A 2 (Agency: Control target vs. Control distractor) × 4 (Delay interval: 0 s, 0.5 s, 1 s, 1.5 s) repeated-measures ANOVA on visual search response time revealed no significant main effect of delay ($F(3, 90) = 1.66$, $p = .18$, $\eta^2 = .052$), indicating that temporal disruptions between the Control and Visual Search Phases did not significantly impair performance. However, we found a significant main effect of agency ($F(1, 30) = 12.58$, $p < .001$, $\eta^2 = .30$, see Figure 4), with participants responding faster to targets they had previously controlled. The agency-delay interaction was not significant ($F(3, 90) = .10$, $p = .95$, $\eta^2 = .003$). These results addressed RQ2 by demonstrating that the beneficial effect of agency on visual search persists robustly across temporal delays of up to at least 1.5 seconds. Further research is needed to establish the upper temporal boundary of this effect.

To evaluate the influence of search target location on the agency effect, we conducted multiple paired-sample t-tests, applying Holm's sequential Bonferroni correction [38]. As the agency effect did not vary across delay intervals, the data were combined across intervals for analysis. The results showed that when the target appeared in the bottom-right corner (closest to the actual hand), participants initiated responses faster for previously controlled versus non-controlled targets (controlled: 1167 ± 32 ms; non-controlled: 1282 ± 36 ms; $t(30) = -2.99$, $p_{corr} = .022$, Cohen's $d = .54$). For search targets appeared in the top-left corner (farthest from the actual hand), the agency effect was marginally significant ($t(30) = -1.99$, $p_{corr} = .059$, Cohen's $d = .36$), with mean response times of 1303 ± 55 ms for controlled versus 1388 ± 48 ms for non-controlled targets. This suggests that the effect was attenuated when the search target was located beyond peripersonal space. Moreover, we found that the overall response time was significantly slower when the target was in the farthest position (Top-left quadrant: 1344 ± 47 ms) compared to the closest position (Bottom-right quadrant: 1223 ± 28 ms, $t(30) = 2.34$, $p = .026$) (see Figure 5).

6 Experiment 3: Spatial Adaptability

The ability to integrate multiple visual perspectives is critical in procedures such as laparoscopic surgery. A surgeon must continually switch between monitor views to assess a target site (e.g., a tumor) while also tracking previously dissected anatomical structures, like nerves, from differing vantage points. Experiment 3 assessed the

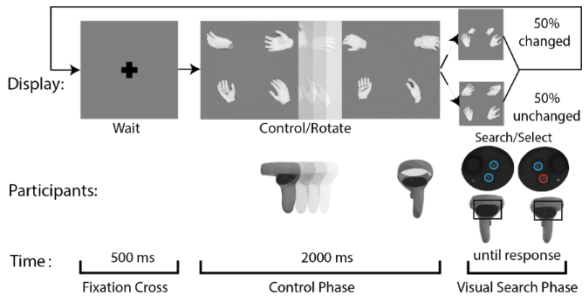

Figure 6: Illustration of a single trial in Experiment 3. Participants were informed before the experiment that the viewing perspective may change in the Visual Search Phase.

spatial adaptability of agency by rotating the stimulus layout by 90° between control and search phases. This design helps to evaluate whether agency effects transfer across changes in viewing perspectives and spatial configuration.

6.1 Participants

We employed a procedure similar to that used in previous experiments to estimate the sample size. For a 2 (Agency: Control target vs. Control distractor) × 2 (Spatial transformation: Changed vs. Unchanged) repeated-measures design, we anticipated a minimum sample size of 30 to detect a medium effect size (f = .25, η^2 = .06), with an alpha level of 0.05 and a power of 0.9. The same cohort of 31 participants from Experiment 2 took part in Experiment 3.

6.2 Stimuli, Procedure and Apparatus

We altered the presentation of stimuli in the Visual Search Phase from a frontal view (i.e., unchanged condition) to a perspective view (i.e., changed condition) for half of the time, requiring observers to rely not solely on 2D positional cues, but also on a 3D mental representation of the object to conduct the search task (see Figure 6). All parameters were identical to those in Experiment 2, except for the layout-altered condition in which we adjusted the relative size of the closer virtual hands to 35° × 34° and the more distant hands to 15.95° × 16.12°. Experiment 3 included 106 trials: 10 practice trials and 96 experimental trials. The two viewing perspective conditions were randomly assigned, each appearing 50% of the time. Participants completed the entire experiment in approximately ten minutes. Experiment 3 employed the same apparatus as Experiment 2.

6.3 Data Analysis

We excluded incorrect trials and outliers (response times less than 30 ms or greater than 3000 ms) from the data analysis which comprised 3.1% of the data (1.9% errors, 1.2% outliers). Similar to previous experiments, participants maintained a high and consistent level of accuracy (>90%) across all conditions. Therefore, our analysis focused primarily on reaction time for the visual search task.

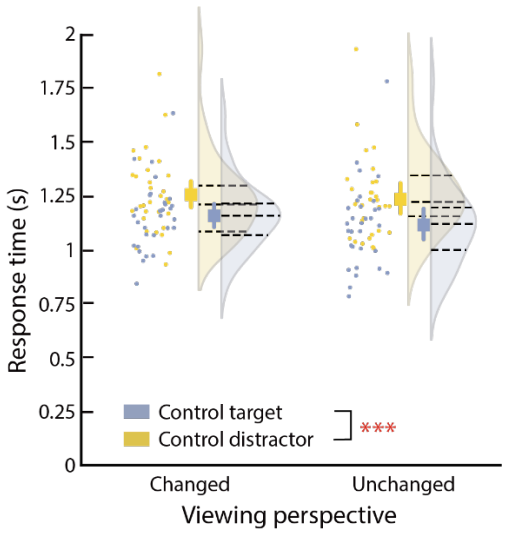

Figure 7: Across both viewing perspectives, prior control of a target led to consistently faster identification compared to targets that were not previously controlled. ***, p < .001

6.4 Results and Discussion

A 2 (Agency: Control target vs. Control distractor) × 2 (Spatial transformation: Changed vs. Unchanged viewing perspectives) repeated-measures ANOVA on the response time revealed significant main effects of both spatial transformation (F(1, 30) = 5.25, p = .029, η^2 = .15) and agency (F(1, 30) = 16.50, p < .001, η^2 = .35, see Figure 7). The interaction between them was not significant (F(1, 30) = .12, p = .73, η^2 = .004). These results indicate that while spatial transformation slowed down the overall response time, the enhancing effect of agency on visual search remained robust across the two viewing perspective conditions. Our findings align with the predictive processing frameworks such as pre-activation and active inference theories [39-41], indicating that prior control enhances the recognition of the controlled object even when its spatial context changes (RQ3). In practical term, a surgeon's prior instrumental control of an anatomical structure could enhance his ability to identify it for many other similar structures (i.e., distractors) on a monitor, even after a shift in viewing angle, thereby helping to maintain precision and situational awareness during a procedure. While this effect may seem intuitive, it offers a valuable insight for human-computer interaction design. For instance, granting a user simple control, such as a click or drag command, over a critical interface element could facilitate their later performance in locating or interacting with that target amidst visual complexity.

We also conducted multiple paired-sample t-tests with Holm's sequential Bonferroni correction [38] to evaluate the effect of search target location. The agency effect was significant in all four quadrant (t(30) ≥ 2.35, p ≤ .034, Cohen's d ≥.42). Furthermore, similar to the results in Experiment 2, the overall response time was longer when the target position was farthest (Top-left quadrant: 1295 ± 40 ms) than when it was closest to real hand (Bottom-right quadrant: 1155 ± 27 ms, t(30) = 3.58, p = .0012, see Figure 8).

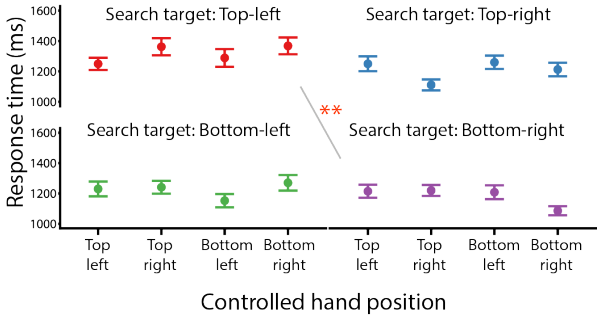

Figure 8: Illustration of response times as a function of search target position and the controlled hand position in four quadrants in Experiment 3. **, p < .01

Furthermore, we found that while changes in viewing perspective (and the resulting repositioning of the virtual hands) modestly increased overall response times, the performance benefit conferred by agency remained intact. Critically, the absence of an interaction effect indicates that faster responses to agency-associated targets were maintained irrespective of spatial rearrangement. This suggests that agency facilitates visual search through a spatially flexible mechanism, likely involving the cognitive remapping of the controlled object, rather than through localized attentional persistence or gaze facilitation. Notably, this agency advantage was observed even when the target appeared in the position farthest from the hand, a condition that generally produced the slowest responses. This further underscores the adaptability of agency-driven attention across diverse spatial configurations.

7 General Discussion

Our results demonstrate that the sense of agency, induced by prior active, self-generated control over a stimulus, guides attention in a robust and persistent manner in VR environment. Specifically, the resulting attentional benefit remained robust against physically salient distractors (RQ1), persisted for temporal delays of at least 1.5 seconds (RQ2), and resisted changes in observational perspective (RQ3).

In Experiment 1, we found that while an invalid exogenous cue slowed overall search, the facilitatory effect of agency remained intact. Crucially, these two effects, the cost of distraction and the benefit of agency, coexisted independently. Building on this, Experiments 2 and 3 examined whether the agency effect could be undermined by temporal or spatial disruptions. We found that the processing advantage for a self-controlled target endured a delay of at least 1.5 seconds between the control and search phases. Furthermore, the benefit was maintained despite spatial transformations and shifts in viewing perspective. Collectively, these findings reveal that agency-driven attention exhibits strong temporal persistence, spatial adaptability, and resilience against external attentional competition.

However, we also identify key limitations for agency-based interfaces. Importantly, agency-driven attention, while robust, functions in parallel with bottom-up salience and does not supersede it. This clarifies a central design implication: while agency provides a powerful attentional scaffold, it must be complemented by other cues (e.g., visual highlighting) in contexts of extreme spatial separation or competing high-salience stimuli.

7.1 Underlying Mechanisms of Agency

There are a few alternative theories that could help explain the facilitatory effect of agency on visual search performance, which we tentatively included in the discussion. Firstly, we have revealed that agency serves as a natural guidance for attention during interaction in an environment contains multiple agents. To examine the persistence and adaptability of agency effects, we differentiate between body agency, defined as the sensation of controlling one's own body, and external agency, which refers to the perception of controlling an external object [42]. We hypothesize that the lack of detrimental effects due to delays and changes in stimuli layout on agency may be explained by the inherently tolerant characteristics of external agency. For example, considerable delays can significantly impair the perception of body agency, whereas external agency demonstrates a higher temporal tolerance [43]. To illustrate, prior research has often focused on the simple delay between a single action and its immediate effect. In human-computer interaction, however, a single user action can trigger a cascade of multiple outcomes. For example, pressing a button on a UI interface might cause the button to change color and subsequently alter application settings. In our study, the precise, real-time synchronization between the participant's wrist rotation and the corresponding avatar movement fostered a robust sense of agency. This strong sense of control, in turn, enhanced the transferability of agency-driven attention to subsequent tasks.

The perception of agency concerning the modulation of an external object demonstrates a greater persistence over prolonged temporal interval and spatial transformations in comparison to body agency. Additionally, the sense of agency established through active control facilitates stimulus detection by lowering its information processing threshold. This reflects a neural prioritization to synthesize action outcomes, which selectively enhances attentional allocation to self-caused events.

7.2 Effects of Peripersonal Space

The brain gives privileged, multisensory processing to stimuli and interactions occurring near the body, resulting in faster and more robust neural and behavioral responses. This region of enhanced processing is commonly referred to as peripersonal space, although its precise boundaries differ across studies [44]. Peripersonal space could shorten the reaction time to a given stimulus, and this effect was expected to decrease with the increase of distance [45]. For instance, patients with left visual field deficit showed improvement on a visual task of identifying a number, specifically when fingers were visibly presented close to the number [46]. This improvement of visual task performance is not a result of attentional shifting by changing focus, rather, it is a direct attentional prioritization benefit brought by the presence of hand [47]. A pre-activation model offers a unifying account, proposing that the brain enhances perceptual processing of self-generated stimuli, reducing prediction error and thereby decreasing perceived surprise while increasing

detection speed, even as discriminability may decline [41]. This framework reconciles the divergent results by reframing them as complementary manifestations of agency's influence on perception.

In our study, by comparing the response time of target appearing at Top-left (farthest away from the real-world hand) vs. Bottom-right (closest to real-world hand), we found generally slower response when target was presented at a farther position to observer's real hand, which was consistent with the notion of behavioral response enhancement decreases with proximity to one's body, hand specifically [45].

7.3 The Role of Planned Voluntary Actions in Agency

Beyond physical proximity, voluntary action is another key factor that enhances behavioral performance. The sense of agency that accompanies voluntary actions appears to be critical. For instance, one study found that merely seeing one's own hand on a monitor did not improve motor response speed. However, performance was enhanced when the displayed hand performed a voluntary action, indicating that the sense of agency itself contributes to faster behavioral responses [48]. Consistent with this view, our study found that only the virtual hand mirroring the participant's real movement generated a sense of agency, which in turn led to its prioritized processing during a subsequent visual search task among distractors.

Nevertheless, the facilitatory effect of agency was likely to be attenuated by the spatial context of the planned voluntary action. This aligns with Action Field Theory, which posits that peripersonal space is dynamically shaped by stimuli interacting with moving body parts [49]. Our results refine this view. In Experiment 2, we found only a marginally significant agency benefit when the target appeared in the top-left quadrant, the position farthest from the hand. However, in Experiment 3, this spatial disadvantage over the facilitatory effect was mitigated. Given the absence of a delay between the Control and Visual Search Phase in Experiment 3, we postulate that the attenuation of agency effects with increasing distance from the actual hand might be partially attributable to the temporal contingency of the action effect. A dynamic model based on Action Field Theory offers an alternative explanation: the persistent agency effect, even when perspective shifts increased the target's virtual distance, endured because the relevant spatial relationship was interpreted within the 2D plane of the display. According to this model, the perceived distance between control and target locations remained close (see Figure 9), maintaining the facilitatory effect.

8 Implications and Future Work

8.1 Design Technology

A primary contribution of this work lies in its demonstration of the robustness of agency-driven attention, particularly its resilience to attentional noise (RQ1), temporal delays (RQ2), and spatial transformations (RQ3). These findings characterize the agency effect as both durable and adaptable across dynamic task conditions. Several important implications arise from these results.

Firstly, agency exhibits resilience to external distractions; however, it cannot eliminate their adverse effects. In designing visual guidance systems, it is recommended that designers minimize inconsistencies between visual cues and agency, promoting synergistic integration to optimize attentional direction. Selective attention remains intact during the transition between disparate displays, once the sense of agency has been intactly formed. In surgical applications, for instance, a surgeon may operate while viewing anatomical structures through an augmented reality display, alternating focus between the virtual overlay and the physical patient. It is anticipated that, despite changes in perspective and system lag, attentional prioritization will persist for the surgeon-controlled virtual element compared to a perceptually identical but non-controlled element.

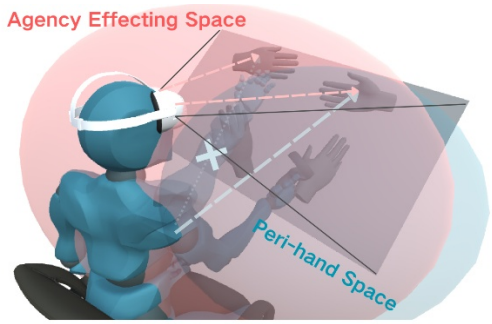

Figure 9: Illustration of proposed dynamic action field and affected areas.

Action affordance serves as an additional determinant of agency effects beyond proximity to the body [50]. The facilitation effect of agency on visual search weakened beyond the operator's peripersonal space, resulting in increased response times to search for target stimuli located at greater distances away from the operator's real hand. Nevertheless, modifying the visual perspective to reduce the motoric effort required for potential actions, such as favoring vertical, directly reachable arm movements over oblique or upward reaches, enhances neural encoding of action possibilities. This adjustment effectively alleviates spatial restrictions, even at increased distances from the body. In virtual hand representations within user interfaces, positional considerations should aim to minimize the biomechanical effort required for actions, aligning virtual spatial arrangements with natural motoric efficiencies.

8.2 Future Work and Limitations

First, although we semi-randomized the amplitude and temporal onset of distractor movements, their potential influence on agency detection remains unquantified. Future work should explicitly analyze how movement kinematics affect agency attribution. Second, our agency manipulation was binary: participants either fully controlled a virtual hand or had no control. This design does not capture more nuanced, real-world scenarios involving partial or mixed controllability. Future study should identify the threshold of control required for agency emergence, particularly in shared-control or co-embodiment contexts [51]. Third, although we controlled for hand-environment spatial relationships, we did not parametrically vary avatar position or environmental context. Systematic manipulation of these factors would clarify how action affordances and

spatial congruence modulate agency-driven attention. Finally, technological constraints limited our use of high-precision eye-tracking technique to dissect attentional mechanisms. As the relationship between gaze and attentional shifts in VR remains debated [47], future studies should integrate eye-tracking under carefully controlled conditions to validate and extend our findings. These directions would strengthen the ecological and theoretical validity of agency research while informing more robust virtual interaction design.

9 Conclusion

Through three controlled experiments, using visual search performance as a behavioral proxy to measure the implicit effects of agency, we confirm the robustness of agency-driven attentional guidance. The benefit withstands competing distractors, lasts over a second, and transfers across different observational perspectives. Furthermore, when a delay was introduced between the control and subsequent visual search, the agency effect attenuated for targets presented beyond the operators' peripersonal space. Collectively, our findings demonstrate that self-generated active control of a stimulus induces a sense of agency, which exerts an independent and generalizable influence on visual attention toward that stimulus. The results offer practical insights for interactive system design, particularly in virtual reality and game development, by providing evidence-based strategies to sustain user engagement through the intentional use of agency-driven attentional guidance.

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